

Algorithms

1: User Registration

```
BEGIN
  IF email already exists THEN
    RETURN "Email already registered"
  ENDIF
  Hash password
  Create user record
  Send verification notification
END
```

2: Check Field Availability

```
BEGIN
  Fetch slots for selected
  field and date
  FOR each slot
    IF overlapping AND not
    available THEN
      RETURN "Not
      Available"
    ENDIF
  ENDFOR
  RETURN "Available"
END
```

3: Create Booking

```
BEGIN
  IF slot is not available THEN
    RETURN "Slot already booked"
  ENDIF
  Create booking with status Created
  Set slot unavailable
  Update booking to Reserved
  Send notification
END
```

4: Process Payment

```
BEGIN
  Initiate payment
  IF payment successful THEN
    Save payment record
    Set booking status to Paid
  ELSE
    Cancel booking
    Release slot
  ENDIF
END
```

5: Dynamic Pricing

```
BEGIN
  Set multiplier = 1.0
  IF demand is High THEN
    multiplier = 1.5
  ELSE IF demand is Medium THEN
    multiplier = 1.2
  ENDIF
  IF time is Peak THEN multiplier
  += 0.3
  Final price = base price *
  multiplier
END
```

6: Group Matchmaking

```
BEGIN
  Search for open group
  IF group found THEN
    Add player to group
  ELSE
    Create new group
  ENDIF
END
```